



Introducing an optimum gas turbine inlet temperature (TIT) based on a 4E analysis: a case study of northeastern Iran

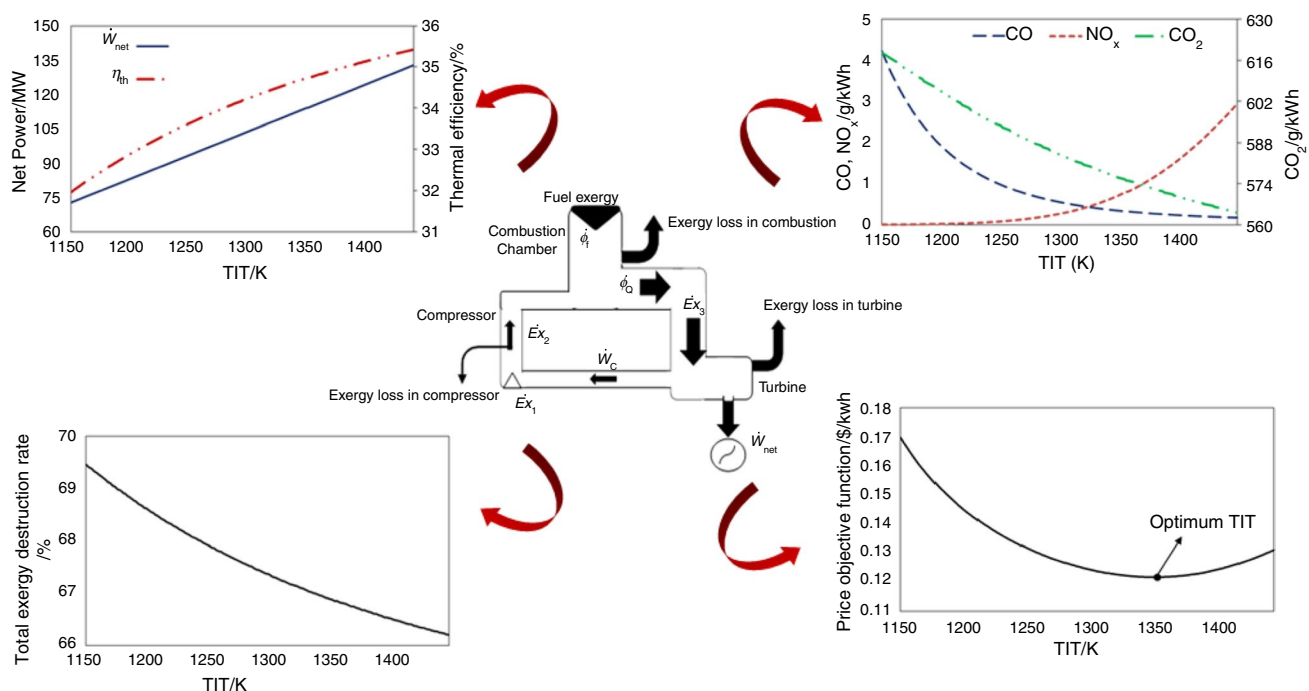
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Abstract

Determining the maximum temperature of gas turbine is one of the challenges in energy conversion to achieve the suitable performance of gas turbine systems. For this purpose, based on the energy, exergy, environmental, and economic (4E) analyses, the effects of changing turbine inlet temperature (TIT) on a gas turbine power plant in northeastern Iran were studied. The results showed that increasing TIT enhanced net power and efficiency, so that increasing TIT about 10 K enhanced net power by 1.7%. Of course, on the other side, higher TIT increases the NO_x emissions and the cost of materials and alloys. Therefore, the results indicated that higher TIT may not be necessarily more suitable. To find an optimal temperature, a price objective function was introduced in which the costs of power generation, net power, efficiency, and environmental taxes were effective. Finally, for the investigated power plant, the optimum TIT of 1328 K was introduced in which the cost of power generation could be minimized by about 0.115 \$/kWh.

Graphic abstract



Keywords 4E analysis · Gas turbine · Turbine inlet temperature · Optimization

Introduction

One of the highly popular power generation systems in the industry is a gas turbine. However, improving the performance of gas turbines is very important [1]. Many studies were carried out on enhancing the net power and improving the efficiency of gas turbines [2–4]. Increasing turbine inlet temperature (TIT) is a way for achieving the above-mentioned purposes [5, 6].

The impacts of increasing TIT on the performance of gas turbines were surveyed in several studies [7, 8]. Athari et al. [9, 10] showed that increasing TIT by about 11 K enhanced both energy and exergy efficiencies by 0.45. Rahman et al. [11] also indicated that increasing TIT by 200 K enhanced the thermal efficiency by 3%. Moreover, in Nigeria, Oyedepo et al. [12] studied a gas turbine power plant. They revealed that increasing TIT by 200 K reduced the cost of exergy destruction by about 29%. Furthermore, the results displayed that the increment of TIT reduced the emissions of CO₂ and decreased the cost of its environmental impact. In addition, Ebadi and Gorji-Bandpy [13] carried out the exergy analysis of a 116-MW gas turbine power plant based on the variations of TIT. The results demonstrated that increasing TIT by 10% enhanced the exergy efficiency by 18% and decreased the exergy destruction by 12%. In a similar vein, Al-Doori [14] showed that increasing TIT by 120% reduced the total exergy destruction of the plant by 23.7%. Additionally, Sahu and Sanjay [15, 16] compared a simple turbine with a reheat gas turbine carrying out an exergoeconomic analysis. The results illustrated that a higher TIT increased specific work of power plant (kJ/kg) in both cases. Also, some studies investigated the effects of increasing TIT on the performance of integrated cycles with a gas turbine [17, 18]. Choudhary and Sanjay [19] designed an integrated solid oxide fuel cell with an intercooled air-film blade-cooled gas turbine. The results showed that increasing TIT enhanced the specific work of a hybrid plant. The authors indicated that the hybrid cycle was optimum at TIT = 1800 K in which the entropy generation was minimized.

Although increasing TIT gives a higher net power and thermal efficiency [20], there are some practical difficulties for increasing the TIT. The main problem is the material property limitation. Thus, due to metallurgical considerations, TIT cannot be assumed to be extremely high. Therefore, in order to prevent the damage of turbine blades, the maximum value of TIT is restricted. Of course, this temperature can be raised by cooling the turbine components or using special high-temperature materials, which are expensive processes. [5, 21–24].

Generally, due to the high temperature of combustion in the gas turbines, thermal NO_x is often generated [25, 26]. Therefore, the increment in the NO_x emission is another

problem of increasing TIT [27, 28]. With regard to the Kyoto Protocol [29] and Paris Agreement [30], in order to decrease the emission of air pollutants and prevent global warming, decreasing NO_x emission is highly important. Researchers' efforts for designing industrial gas turbines with the lowest NO_x emissions approved more stringent regulations for generating power. At 15% oxygen, NO_x emissions below 50 mg/m³ (25 ppm) are essential due to the European Industrial Emissions Directive (IED), which came into force in 2016 [31]. In many countries, NO_x emissions less than 10 ppm are needed. For example, in Italy, the NO_x emissions below 10 ppm are regulated [31]. The combustion temperature directly affects the NO_x emission, while a lower combustion temperature results in lower NO_x emissions [32]. Andrews [33] showed that at different primary zone temperatures, including 1700, 1800, and 1900 K, the values of NO_x emissions were 1, 2, and 3.5 ppm, respectively.

In the previous literature, the effects of various TITs on the net power, efficiency, and environmental pollutants were individually investigated. Considering previous studies, increasing TIT has two contradictory effects on the gas turbine cycle. Increasing TIT improved the net power and efficiency [5–10] that could be considered as positive effects on the cycle, whereas it might also increase NO_x, and such an increase could be considered as a negative impact [27, 28]. Therefore, based on the energy, exergy, environmental, and economic analyses (4E), the effects of various TITs on the performance of a power plant in northeastern Iran were investigated. In order to evaluate the influences of various TITs, a thermo-environmental-economic (4E) model was developed and the total price of electricity generation consisting of initial investment, operation and maintenance cost, fuel cost, and environmental taxes was found. With the aim of making a quantitative comparison between different values of TIT, based on the developed model, a price objective function (POF) was determined. Finally, considering the presented POF, an optimum TIT was introduced.

In summary, some of the main novelties and motivations of this paper are as follows:

- 4E analysis of a gas turbine in details.
- Evaluating the impacts of maximum cycle temperature (turbine inlet temperature) on the thermodynamic, environmental, and economic parameters simultaneously.
- Introducing a comprehensive optimum value of turbine inlet temperature in which the cost of power generation is minimum.

Power plant description

In the present study, a gas turbine power plant in northeastern Iran, which had six gas turbine units with 159 MW nominal capacity for each one, was investigated. The recorded data of the power plant are listed in Table 1. Natural gas was used as fuel supplied by Shahid Hashemi-Nezhad refinery in the northeast of Iran. Also, the chemical composition of fuel is indicated in Table 2.

Figure 1 illustrates a schematic of a gas turbine. Fresh air at the ambient pressure and temperature entered the compressor for compression. The compressed air and fuel were burnt in the combustion chamber, and the hot combusted gases expanded in a turbine for generating power.

Governing equations

Changing TIT (T_3) affects the gas turbine performance, the emission of environmental pollutants, and the cost of power generation. Therefore, a comprehensive analysis is needed to investigate the effect of changing TIT on these parameters. 4E analysis is a simultaneous evaluation method based on the energy, exergy, environmental, and economic (4E) analyses. This analysis can be viewed as a comprehensive analysis that was used in the current study.

Table 1 Input data used in this paper [34]

Description	Symbol	Sample value
Altitude above the sea level		1097/m
Ambient relative humidity	RH	10%
Ambient temperature	T_1	311/K
Compressor inlet mass flow rate	$\dot{m}_a$	409/kg s ⁻¹
Pressure ratio of compressor	r_c	11.7
Compressor isentropic efficiency	η_c	0.86
Turbine inlet temperature	T_3 (TIT)	1370/K
Turbine isentropic efficiency	η_t	0.89
Net power	$\dot{W}_{net}$	118.8/MW

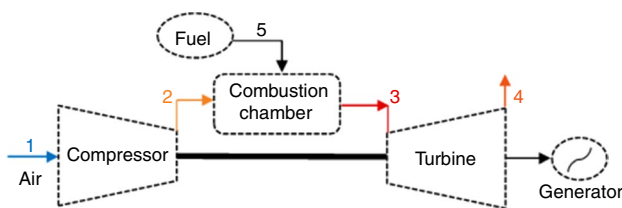


Fig. 1 A configuration of the gas turbine

Table 2 Chemical composition of natural gas from the Shahid Hashemi-Nezhad refinery [35]

Component	Chemical formula	Percent mole fraction
Carbon dioxide	CO ₂	0.055
Nitrogen	N ₂	0.428
Methane	CH ₄	98.640
Ethane	C ₂ H ₆	0.593
Propane	C ₃ H ₈	0.065
Isobutane	C ₄ H ₁₀	0.015
Normal butane	C ₄ H ₁₀	0.034
Isopentane	C ₅ H ₁₂	0.026
C6+	C6+	0.125
		Total = 100%

Energy analysis

Initially, the mass and energy balances were applied to all components for steady-state conditions. Of course, it should be noted that the kinetic and potential energy terms were negligible [34, 36]. Based on Fig. 1, the compressor was the first component. The compressor outlet air conditions could be calculated as the following:

$$P_2 = r_c \times P_1 \quad (1)$$

$$h_2 = h_1 + \left[\frac{h_{2s} - h_1}{\eta_c} \right] \quad (2)$$

where r_c and η_c are the compression ratio and isentropic efficiency of compressor that were assumed to be 11.7 and 0.86, respectively.

Therefore, based on the energy balance, the work consumed by the compressor was calculated as follows:

$$\dot{W}_c = \dot{m}_a h_2 - \dot{m}_a h_1 \quad (3)$$

where $\dot{m}_a$ is the mass flow rate of the compressor (kg/s).

The released energy by the combustion was obtained utilizing the lower heating value of fuel (LHV (kJ/kg)) as follows:

$$\dot{Q}_H = \dot{m}_f \times \text{LHV} \quad (4)$$

$\dot{m}_f$ is the fuel mass flow rate (kg/s) and LHV is assumed to be about 50,010 kJ/kg [36].

The energy balance in the combustion chamber was calculated as:

$$\dot{m}_a h_2 + \dot{m}_f \text{LHV} \eta_{cp} = (\dot{m}_a + \dot{m}_f) h_3 \quad (5)$$

Here, the combustion process efficiency (η_{cp}) is considered to be 99% [37] and h_3 is the combustion chamber outlet enthalpy (kJ/kg).

The turbine outlet gas conditions could be calculated as follows:

$$h_4 = h_3 - [\eta_t \times (h_3 - h_{4s})] \quad (6)$$

Here, η_t is the isentropic efficiency of turbine and h_{4s} is the outlet enthalpy of turbine in the isentropic conditions.

The output power of turbine by a control volume around the turbine was calculated based on the following equation:

$$\dot{W}_t = (\dot{m}_a + \dot{m}_f)(h_3 - h_4) \quad (7)$$

The net power output of system was expressed as follows:

$$\dot{W}_{\text{net}} = \dot{W}_t - \dot{W}_c - \dot{W}_{\text{fuel injection}} \quad (8)$$

Here, $\dot{W}_{\text{fuel injection}}$ is the consumed power for injecting fuel ($\dot{W}_{\text{fuel injection}} = \int \eta_p v dp$, $\eta_p = 0.85$).

Specific fuel consumption (SFC) was defined by Eq. (9).

$$\text{SFC} = \frac{3600 \times \dot{m}_{\text{fuel}} \times 1000}{\dot{W}_{\text{net}}} \quad (9)$$

Eventually, the gas turbine thermal efficiency was obtained as follows:

$$\eta_{\text{th}} = \frac{\dot{W}_{\text{net}}}{\dot{Q}_H} \quad (10)$$

Exergy analysis

Due to irreversible processes, exergy is usually destroyed [38]. A schematic of exergy and exergy loss flow is shown in Fig. 2.

For steady-state conditions, exergy balance was generally determined as follows:

$$\dot{\phi}_f - \dot{W}_{\text{net}} = \sum \dot{E}x_{\text{out}} - \sum \dot{E}x_{\text{in}} + \dot{E}x_{\text{Loss}} \quad (11)$$

$\dot{\phi}_f$ is the chemical exergy of fuel that was obtained by the following equation:

$$\dot{\phi}_f = \dot{m}_f E_f + \dot{W}_{\text{fuel injection}} \quad (12)$$

E_f is chemical exergy of natural gas.

The heat exergy was calculated by Eq. (13).

$$\dot{\phi}_Q = \left[1 - \frac{T_0}{T} \right] \dot{Q}_H \quad (13)$$

The inlet or outlet exergy of components ($\dot{E}x_{\text{in(out)}}$) was obtained by the following equations:

$$\dot{E}x_{\text{in(out)}} = \sum \dot{m}_{\text{in(out)}} [(h_{\text{in(out)}} - h_0) - T_0 (s_{\text{in(out)}} - s_0)] \quad (14)$$

where h_0 , s_0 , and T_0 are the enthalpy, entropy, and temperature at the dead state (ambient conditions in the present study), respectively.

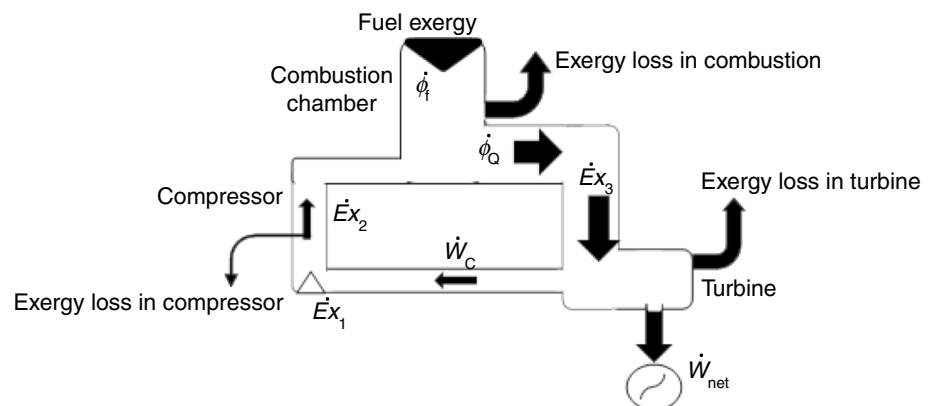
The overall rate of system exergy destruction ($r_{\dot{E}x_{\text{Loss}}}$) could be calculated as Eq. (15):

$$r_{\dot{E}x_{\text{Loss}}} = \frac{\dot{E}x_{\text{Loss}}}{\sum \dot{E}x_{\text{in}}} \quad (15)$$

Environmental analysis

In the power production systems, reducing fuel consumption, increasing the energy and exergy efficiency, and minimizing the impacts of environmental pollutants are the primary purposes. The technology of NO_x control in gas turbine is developed, but gas turbines continue to constitute a very important source of pollutant gases. Increasing TIT also impresses the production of pollutants and raises the NO_x emission. The main emissions produced in any process of combustion are CO_2 , CO, and NO_x . Of these three main emissions, NO_x (nitrogen oxides) has more dangerous effects on human and environment health [39]. In this study, the amount of the pollutant emissions was highly dependent on the combustion technology, fuel type, and operating conditions [40].

Fig. 2 Exergy and exergy loss flow diagram



The amount of the CO and NO_x (g_(CO or NO_x)/kg_{fuel}) in gas turbine was obtained by the following equations [40]:

$$\dot{m}_{\text{NO}_x} = \frac{a_1 \tau \exp(-a_2/T_{\text{pz}})}{P_2^{a_3} (\Delta P/P_2)^{a_4}} \quad (16)$$

$$\dot{m}_{\text{CO}} = \frac{a_1 \exp(a_2/T_{\text{pz}})}{P_2^{a_3} \tau (\Delta P/P_2)^{a_4}} \quad (17)$$

In the above-mentioned equations, P_2 is the inlet pressure of combustion chamber and ΔP is its pressure drop. The residence time in the combustion zone (τ) was considered 0.002 s [40] for this gas turbine model. Also, the adiabatic flame temperature (T_{pz}) and other coefficients of Eqs. (16) and (17) could be defined according to Refs. [40–42].

In the present study, natural gas supplied from Khangiran refinery was considered as consumption fuel. Due to the fact that methane purity in refined natural gas was 99%, the consumption fuel was considered pure methane for simplicity calculations [43]. Thus, the CO₂ emission could be calculated by balancing the carbon between consumed fuel mass flow rate and CO obtained by Eq. (17).

Economic analysis

Considering the main costs of producing power, a price objective function (POF) was presented as follows:

$$\text{POF} = C_I + C_{\text{O\&M}} + C_F + C_P \quad (18)$$

Here, C_I , $C_{\text{O\&M}}$, C_E , and C_P are the initial cost, operation and maintenance (O&M), fuel expense, and environmental taxes, respectively.

The effect of capital cost on the total cost of power production would be computed by the following relation:

$$C_I = \frac{C \cdot I}{t_w \cdot \dot{W}_{\text{net}}} \quad (19)$$

Here, t_w is the total working hour of the plant per year, which was assumed to be 8000 h in this study [34], and C is the initial cost of a gas turbine (\$) that is calculated as follows:

$$C = C_{\text{com}} + C_{\text{cc}} + C_{\text{tur}} \quad (20)$$

where C_{com} , C_{cc} , and C_{tur} are the initial cost of compressor, combustion chamber, and turbine, respectively.

$$C_{\text{com}} = \left(\frac{C_{11} \dot{m}_{\text{air}}}{C_{12} - \eta_{\text{Com}}} \right) \left(\frac{P_2}{P_1} \right) \text{Ln} \left(\frac{P_2}{P_1} \right) \quad (21)$$

$$C_{\text{cc}} = \left(\frac{C_{21} \dot{m}_{\text{air}}}{C_{22} - \frac{P_3}{P_2}} \right) [1 + \text{EXP}(C_{23} T_3 - C_{24})] \quad (22)$$

$$C_{\text{GT}} = \left(\frac{C_{31} \dot{m}_{\text{gas}}}{C_{32} - \eta_{\text{GT}}} \right) \text{Ln} \left(\frac{P_3}{P_4} \right) [1 + \text{EXP}(C_{23} T_3 - C_{34})] \quad (23)$$

Here, the constant coefficients of the above equations are presented in Table 3.

Also the annual interest (I) is calculated as Eq. (24) indicates [42, 45].

$$I = \frac{i(1+i)^L}{(1+i)^L - 1} \quad (24)$$

The system lifetime (L) was normally considered 20 years. The rate of annual interest (i) was assumed to be 9% during the system lifetime or the loan repayment period [46].

Similarly, the impact of annual O&M costs on the produced power was calculated as follows:

$$C_{\text{O\&M}} = \frac{C_{\text{OM}}}{t_w \dot{W}_{\text{net}}} \quad (25)$$

C_{OM} is the O&M cost in \$/year.

The share of fuel cost on the POF would be found as follows:

$$C_F = \frac{\text{Fuel cost} \left(\frac{\$}{\text{kWh}} \right)}{\eta_{\text{th}}} \quad (26)$$

The influence of environmental taxes on the power generation cost was calculated by Eq. (27) [47].

$$C_P = [\dot{m}_{\text{NO}_x} (C_{\text{NO}_x}) + \dot{m}_{\text{CO}} (C_{\text{CO}}) + \dot{m}_{\text{CO}_2} (C_{\text{CO}_2})] \frac{3600}{\dot{W}_{\text{net}}} \quad (27)$$

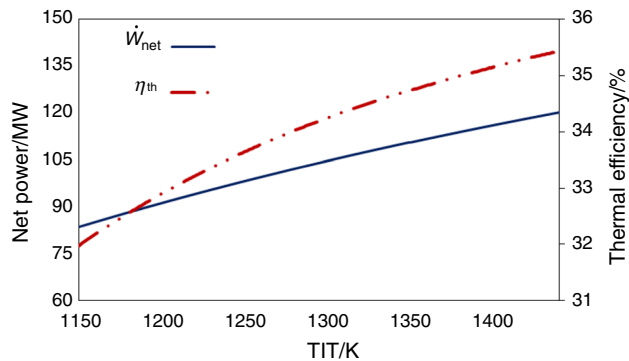
where $\dot{m}_{\text{NO}_x}$, $\dot{m}_{\text{CO}}$, and $\dot{m}_{\text{CO}_2}$ are the exhaust mass flow rates of nitrogen monoxide, carbon monoxide, and carbon dioxide (kg s⁻¹). C_{CO_2} , C_{CO} , and C_{NO_x} are also their external environmental costs (\$ kg⁻¹).

Table 3 Constants used in Eqs. (21–23) for the initial purchase cost of the components [44]

Compressor	$C_{11} = 39.5\$/\text{kg s}^{-1}$	$C_{12} = 0.9$
Combustion chamber	$C_{21} = 25.6\$/\text{kg s}^{-1}$	$C_{22} = 0.995$
Turbine	$C_{31} = 266.3\$/\text{kg s}^{-1}$	$C_{32} = 0.92$

Table 4 The comparison between data of the present study and experimental one

	Experimental data [34]	Modeling data	Error/%
Net power/kW	118.8	118.2	0.5
Thermal efficiency/%	35	34.9	0.28
Turbine outlet temperature/K	808.5	807.2	0.16
Compressor outlet temperature/K	628	665.8	6.01

**Fig. 3** The thermal efficiency and net power versus changing TIT

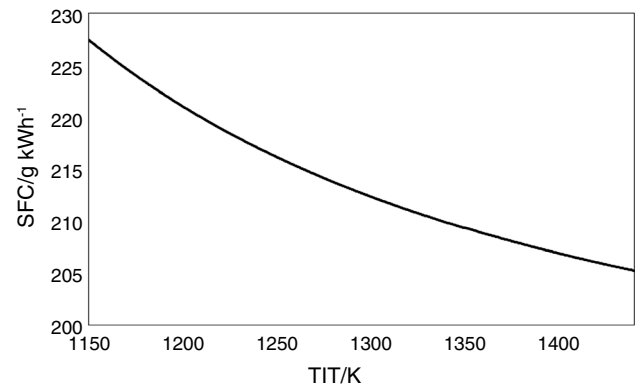
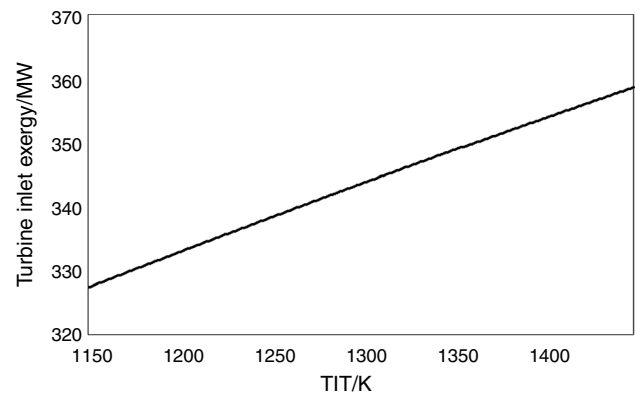
Results and discussion

Regarding the available data in Table 1 recorded from the northeastern Iran power plant, the analyses were performed utilizing the energy, exergy, environmental, and economic equations. The main operating parameters of the system such as net power, thermal efficiency, turbine output temperature, and compressor output temperature were obtained using a thermodynamic simulation. Some experimental data that were recorded in the northeastern Iran power plant are displayed in Table 4. To ensure the accuracy of the proposed model, the data of presented study are compared with that of experimental value in Table 4. The results showed that due to the low error of calculated results, the model validity was confirmed.

In the following section, the effects of increasing TIT on the main parameters are explained.

Energy and exergy analyses

Figure 3 displays the variations of net power and thermal efficiency for different TITs. Higher TIT increased the consumed fuel mass flow rate. Therefore, when the pressure ratio of the compressor was assumed to be constant, the inlet mass flow rate into the compressor was reduced due to the turbine choking and consequently decreased the compressor

**Fig. 4** The SFC versus various TITs**Fig. 5** The turbine inlet exergy versus various TITs

work. Finally, increasing TIT enhanced the net power output as shown in Fig. 3. This point was given in Ref [48]. Regarding Fig. 3, the maximum net power which happened at the highest TIT was 120 MW. In addition, increasing TIT increased the combustion chamber temperature and fuel consumption. Therefore, the rate of heat transfer ($\dot{Q}_H$) was increased. When TIT was raised by 10 K, the net power output ($\dot{W}_{net}$) was increased by 1.7%, whereas the rate of heat transfer ($\dot{Q}_H$) was raised by 1.5%. Eventually, the thermal efficiency of cycle was enhanced. The best efficiency (about 35.42%) was observed at maximum TIT as Fig. 3 indicates. Thus, increasing TIT improved the cycle performance.

In Fig. 4, the variations of SFC versus various TITs are illustrated. At the constant pressure ratio, higher TIT increased the fuel consumption and enhanced the net power output. Of course in this study, the rate of increasing net power is higher than that of fuel consumption. Therefore, it is apparent that the SFC value was decreased by increasing TIT. The lowest SFC was 205.2 g/kWh which happened at the maximum TIT.

The variation of turbine inlet exergy with changes in TIT is illustrated in Fig. 5. The higher TIT increased turbine inlet

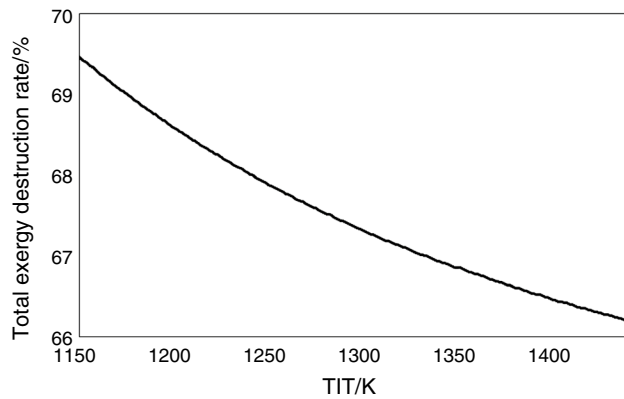


Fig. 6 The exergy destruction versus various TITs

enthalpy and consequently increased turbine inlet exergy. The maximum inlet exergy of turbine was 358.6 MW that happened at the highest TIT. According to Fig. 5, increasing TIT by 10 K increased turbine inlet exergy by 1.1%.

The effects of various TITs on the total exergy destruction are indicated in Fig. 6. Increasing thermal efficiency at the higher TIT decreased the fuel mass flow rate per kWh and reduced the rate of exergy destruction as illustrated in Fig. 6. The minimum exergy destruction was 66.2% that happened at the highest TIT, as displayed in Fig. 6.

From the thermodynamic point of view, increasing TIT enhanced the net power and energy efficiency of the cycle.

Environmental analysis

Changing the temperature affects the amount of environmental pollutants [27, 28]. The effects of increasing TIT on the emission of CO, CO₂, and NO_x are shown in Fig. 7. Increasing TIT increased the combustion chamber temperature and improved the combustion process by reducing CO. Also due to lower SFC at higher TITs, the amount of CO₂ emissions was decreased as indicated in Fig. 7, which was a desirable

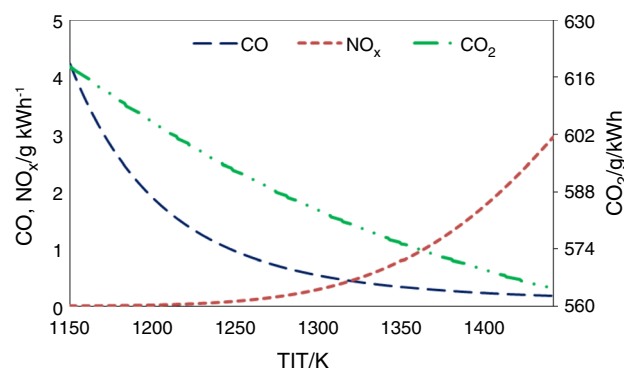


Fig. 7 The emission of pollutants versus various TITs

impact. But, as Eq. (16) indicates, the higher temperature of combustor increased the amount of NO_x emission, which was not favorable. This means that increasing TIT had opposite impacts on the environmental pollutant emissions, so that increasing TIT decreased the positive influences of CO and CO₂ reduction, while it increased the negative impacts of the increment in NO_x.

Considering these two opposite effects of environmental pollutants, there would be an optimum value of TIT in the moderate values.

Economic analysis

Regarding the previous results, increasing TIT had opposite impacts on the different operational factors such as net power, thermal efficiency, rate of exergy destruction, and environmental emissions. Based on the energy and exergy analyses, increasing TIT was always desirable (Figs. 3 and 6). Also, the higher TIT had positive effects on the CO₂ and CO emissions (Fig. 7). However, increasing TIT increased NO_x emission that was a negative impact on the system (Fig. 7). Therefore, a quantitative economical function regarding the total cost of power generation (\$/kWh) was defined that could be calculated by a price objective function (POF) as shown in Eq. (18).

The impacts of capital cost (C_1) were represented on the first term of POF. The higher TIT improved the net power and finally decreased C_1 as shown in Eq. (19). Thus, considering Fig. 8, increasing TIT decreased the amount of C_1 . As was revealed by the negative slope of C_1 , the share of C_1 on the POF was decreased at a higher TIT.

The operation and maintenance cost ($C_{O\&M}$) was the second term of POF, which was estimated at about 29.9 (\$/MWh) [31]. Similar to C_1 , the higher net power decreased $C_{O\&M}$ and eventually the lowest $C_{O\&M}$ happened at the highest TIT as Fig. 9 indicates.

The natural gas price was considered 0.02\$/kWh [46]. The higher TIT enhanced the thermal efficiency, decreased

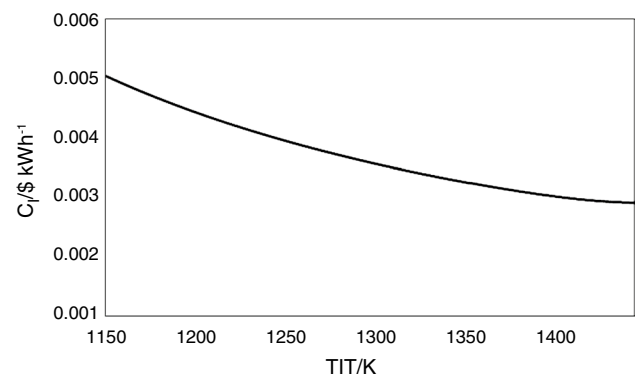


Fig. 8 Changing the capital cost (C_1) versus various TITs

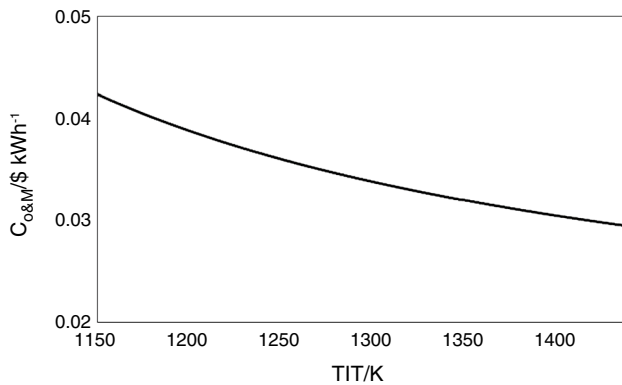


Fig. 9 The operation and maintenance cost ($C_{O\&M}$) versus various TITs



Fig. 10 The cost of fuel (C_F) versus various TITs

Table 5 Rate of environmental taxes [47]

Environmental pollutants	Environmental taxes/\$/kg
NO _x	8.175
CO	6.424
CO ₂	0.024

SFC, and consequently reduced C_F as Eq. (26) shows. Therefore, the higher TIT decreased the cost of the consumed fuel (C_F) as Fig. 10 indicates.

The last term of POF includes the environmental taxes. The value of environmental taxes is presented in Table 5. The higher TIT decreased CO₂ and CO production, while it raised the rate of NO_x generation as shown in Fig. 7. Thus, the variation of TIT affected the environmental cost in two opposite ways. Therefore, regarding Fig. 11, increasing TIT declined the cost of CO₂ and CO, while it exceeded the cost of NO_x. There should be an optimum TIT in the summation

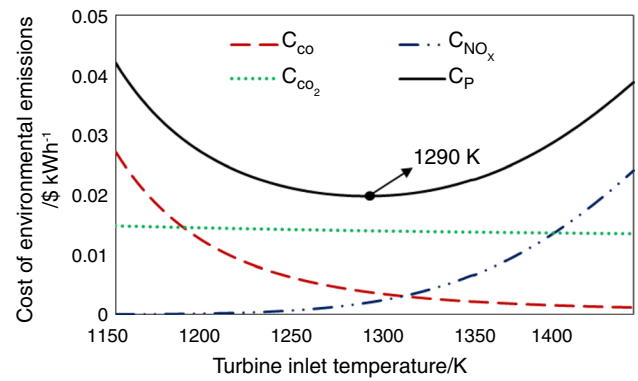


Fig. 11 The environmental taxes versus various TITs

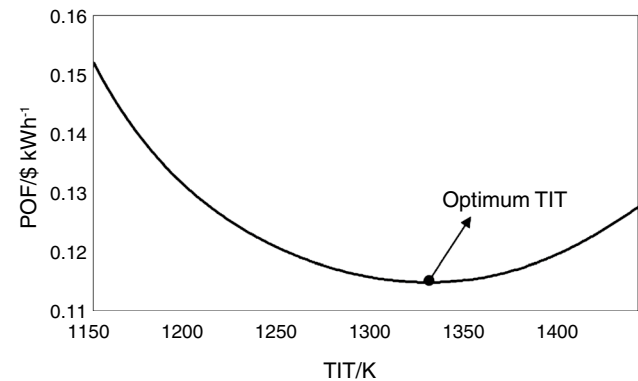


Fig. 12 The variations of the POF versus various TITs

of the environmental taxes. The TIT = 1290 K led to minimum total environmental costs as indicated in Fig. 11.

The effects of various TITs on the different parts of the proposed price objective function are shown in Figs. 8–11. As figures demonstrate, the higher TIT decreased the costs of investment, O&M, and fuel, whereas it had contrary impacts on the environmental pollutant emissions. The total power generation cost (\$/kWh) is displayed in Fig. 12. With regard to the different feedbacks of partial costs toward increasing TIT, there is an optimal value for TIT when the total cost (POF) is minimum as illustrated in Fig. 12. Regarding this figure, in the gas turbine power plant under investigation, the optimum TIT is 1328 K where the POF reaches the lowest value.

The influences of ambient temperature on the optimum TIT

In this part, the impacts of ambient temperature on the optimum TIT are appraised. Increasing ambient temperature decreased the inlet mass flow rate of compressor and net power output [49, 50]. Reducing the net power raised the effect of investment cost (C_I) on the POF. With regard to

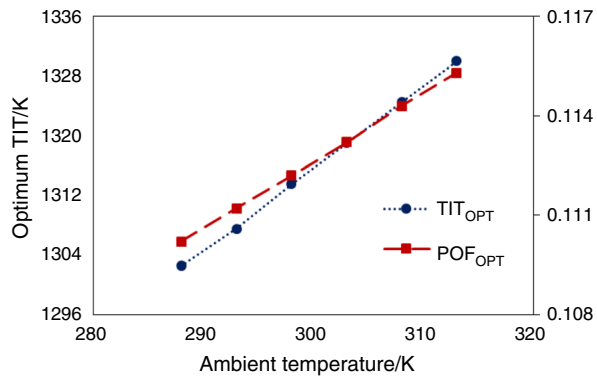


Fig. 13 Changing the ambient temperature versus variations of TIT and POF

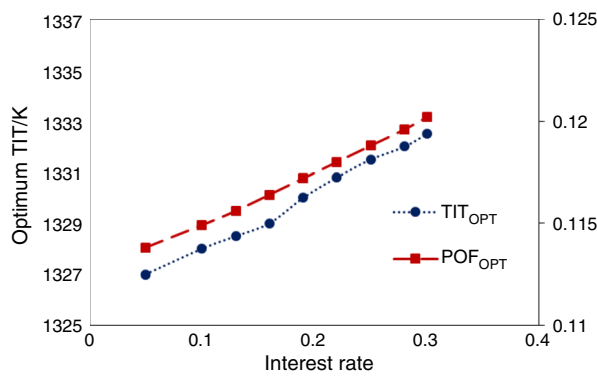


Fig. 14 Changing interest rate versus variations of the optimum TIT and POF

the descending relation of C_f and TIT (Fig. 8), the higher C_f directed the best TIT to the higher amounts as indicated in Fig. 13. For raising the ambient temperature to 5 K, the optimal TIT was increased with an average rate of 0.55% (5 K averagely), while the optimum POF was averagely increased by 1.3%.

The impacts of economic factors on the optimum TIT

The fuel cost, interest rate, and taxes of pollutant emissions are some economic factors that affect the POF and optimum TIT. With regard to the different amounts of these factors in different countries, in the following section, the impacts of economic factors on the optimum TIT are investigated as shown in Figs. 14–16.

In the present study, increasing interest rate increased the first term of POF (C_f) and led to higher optimum POF. Also, raising the share of C_f directed the optimum TIT to higher temperatures as Fig. 14 indicates. As Fig. 14 shows, increasing the interest rate by 2% increased the optimum TIT with an average of 1 K.

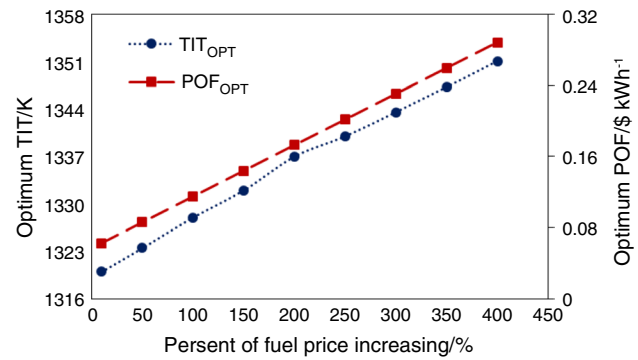


Fig. 15 Changing prices of fuel versus the variations of optimum TIT and POF

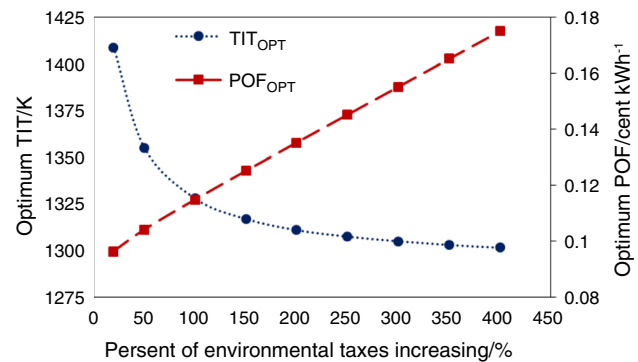


Fig. 16 Changing environmental taxes versus the shift of the optimum TIT and POF

Additionally, the increment of fuel price increased the third term of POF (C_f) and obviously raised the optimum POF as Fig. 15 illustrates. As Fig. 10 reveals, C_f directed the optimum TIT to higher amounts. Therefore, increasing C_f share raised the optimum TIT as indicated in Fig. 15, so that the increment in the fuel prices by 50% increased the optimum TIT by about 4 K averagely.

Changing the pollutant emissions taxes for various TITs is previously displayed in Fig. 11. The total taxes for different values of environmental taxes are indicated in Fig. 16.

Indeed, the higher TIT reduced fuel consumption per kWh of power production and caused better combustion. Eventually, the higher TIT reduced CO and CO₂ emissions as well as their taxes, while, regarding Eq. (16) higher TIT increased NO_x emissions. In other words, the minimum CO and CO₂ taxes happened at the highest TIT, whereas the lowest TIT led to the minimum NO_x tax. Increasing the rate of taxes changed the optimum POF with a constant scale, but the optimum TIT was decreased with a variable slope as shown in Fig. 16. Considering Table 5, NO_x had the highest tax rate and the higher TIT increased NO_x emission. Therefore, the optimum TIT was decreased by increasing the rate of taxes.

Conclusions and recommendations

Turbine inlet temperature (TIT) is one of the effective operational parameters on the performance of gas turbine cycle. In order to evaluate the effects of changing TIT, a gas turbine power plant in the northeastern Iran was selected and based on the energy, exergy, environmental, and economic analyses (4E) was analyzed.

The results indicated that 10 K increase in TIT enhanced net power output by 1.7%. Also, the best thermal efficiency (about 35.42%) occurred at the highest TIT. Concerning the environmental analysis, increasing TIT reduced CO and CO₂ emissions, while the NO_x emissions increased by raising TIT. Due to opposite effects of increasing TIT on the performance of a gas turbine, an important question was raised: what temperature should enter the turbine for optimizing total performance?

For this purpose, considering total costs of power production such as the initial investment, fuel consumption, O&M costs, and environmental taxes, a quantitative economical function was defined for finding the optimum TIT. The results showed that the optimum TIT was 1328 K in the investigated power plant, where the price objective function (POF) was minimum.

With regard to the results, the higher TIT was not necessarily optimal. Therefore, introducing optimum TIT could help to design a more efficient gas turbine, so that it improved the price of power generation and decreased the use of expensive materials.

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